POSTOPERATIVE COURSE VS COMPLICATIONS

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POSTOPERATIVE COURSE VS COMPLICATIONS

NO DISCLOSURES

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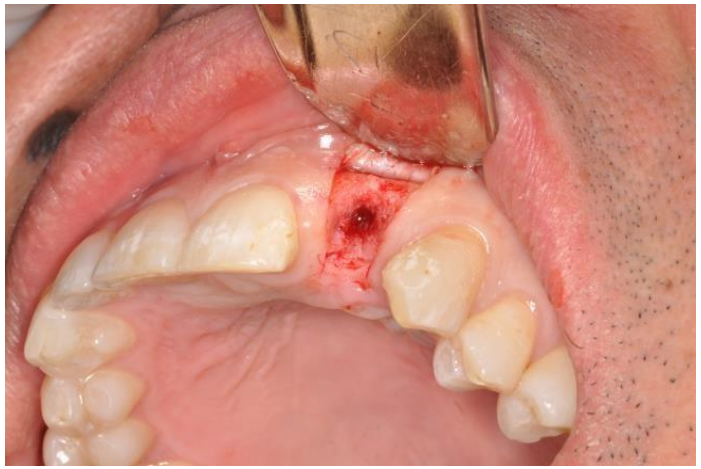
POSTOPERATIVE COURSE

INJURY

Damage (usually acute) to living tissues



WOUND

Injury involving tear of skin or mucosa

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POSTOPERATIVE COURSE

First priority for the body?

Avoid exsanguination!!

Second priority for the body?

Fight invaders - Avoid infection!!

Third priority for the body?

Provide for return to preinjury status (form and function) – wound healing

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TIMELINE

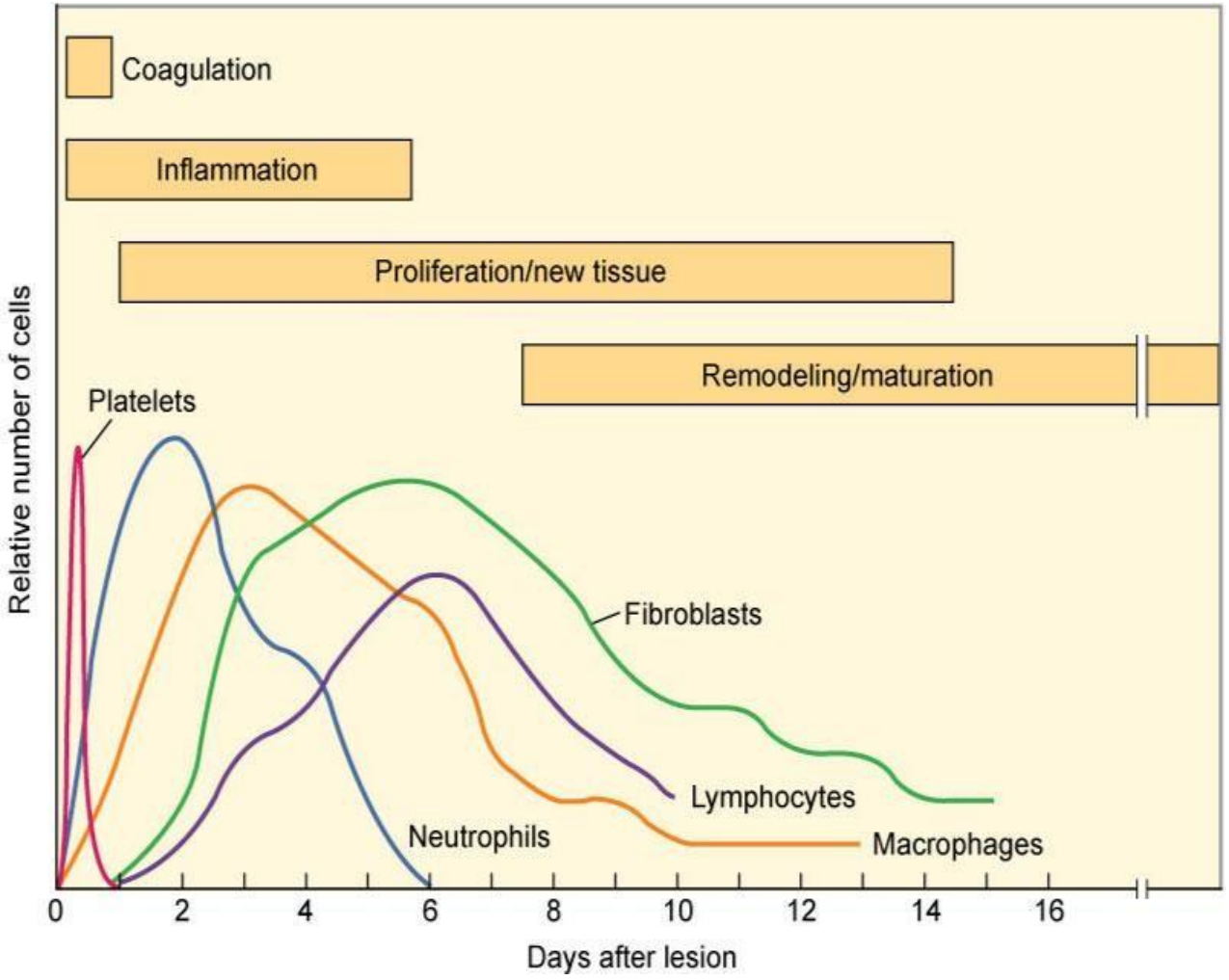
Hemostasis

Inflammation

Proliferation

Inflammatory Phase

Remodeling



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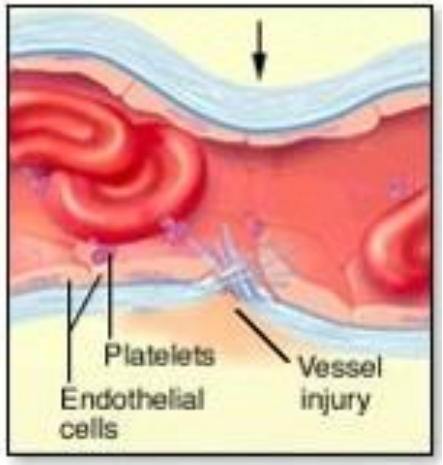
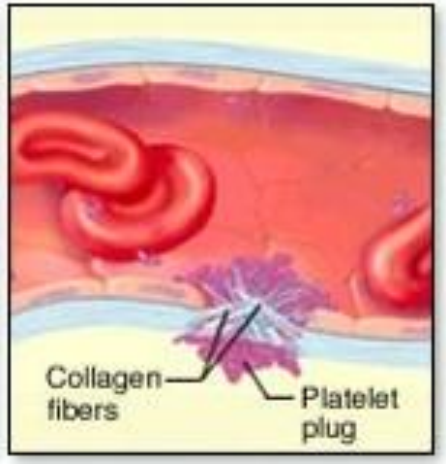
HEMOSTASIS





Vasoconstriction:

Capillary smooth muscle contraction mediated by the mechanical injury and by mediators released by platelets aiming at reducing the blood flow

Platelet activation:

Induced by exposed subendothelial collagen, von Willebrand factor, thrombin. Leads to the formation of the **platelet plug** and release of locally acting mediators

Coagulation:

Initiated by the activation of the coagulation cascade leading to the formation of the **hemostatic plug**

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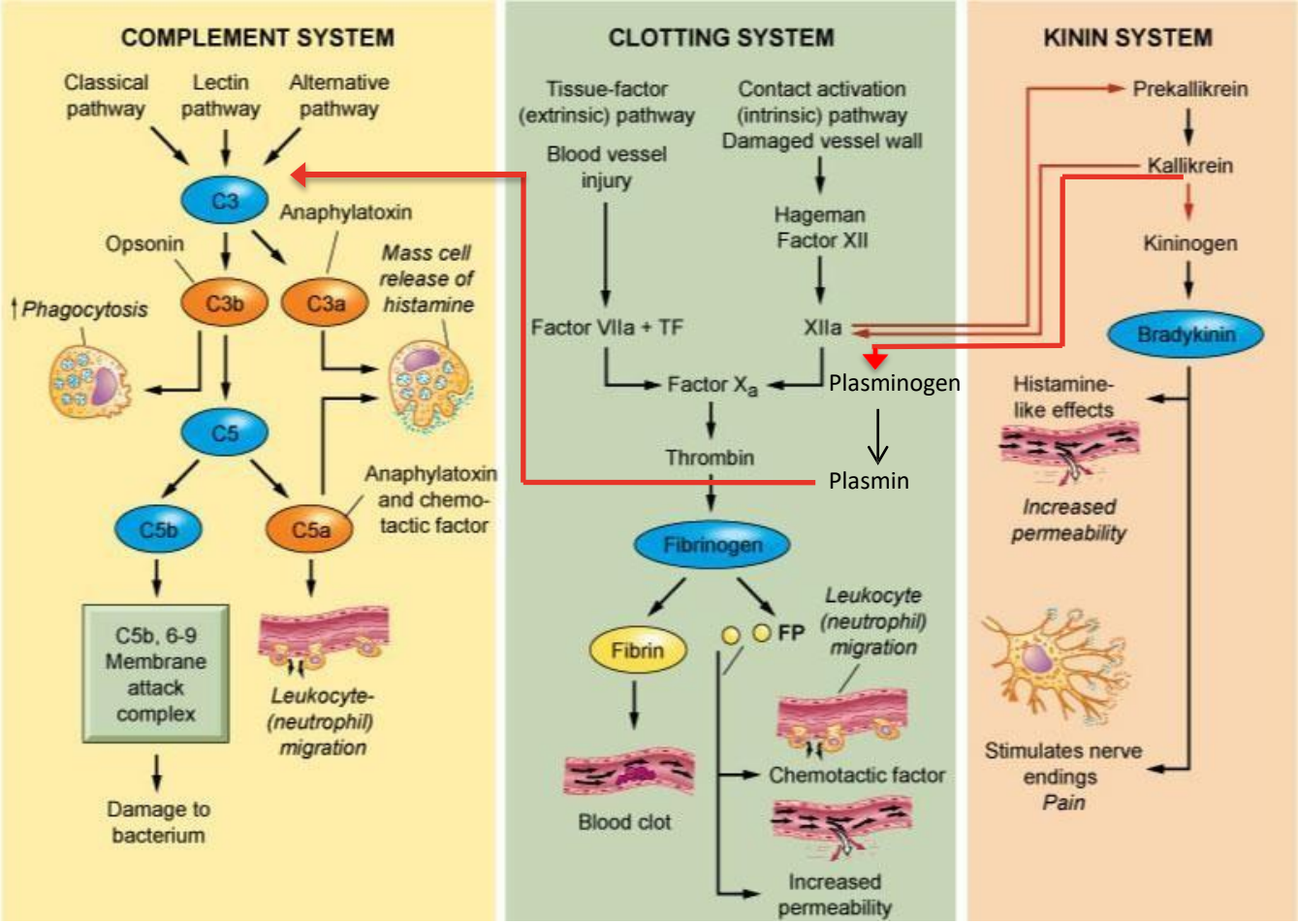
A VERY CLOSELY-KNIT SYSTEM AT WORK



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A VERY CLOSELY-KNIT SYSTEM AT WORK

Tissue Damage

Arachidonic Acid Cascade
(Eicosanoid production)

↓

Exposure of cell membrane phospholipids

↓ phospholipase

Arachidonic Acid Formation

cyclooxygenase (COX) 1,2 ↓ ↓ Lipooxygenase

Prostaglandins
(PGD_2, PGE_2, PGF_2) vasodilation
Prostacyclin (PGI_2) vasodilation, antiplatelet aggregation
Thromboxane A_2 promotes platelet aggregation

Leukotrienes → Vasoconstriction
Bronchospasm
Incr. permeability

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ACTION OF EICOSANOIDS

Prostaglandins:

PGE2 -- vasodilates, intensifies bradykinin induced pain, induces fever
PGD2 -- vasodilates
PGI2 -- vasodilates and is a powerful inhibitor of platelet aggregation TXA2 -- vasoconstricts and stimulates platelet aggregation

Leukotrienes:

LTB4a potent neutrophil and macrophage chemotactic agent
causes aggregation of and increased adherence of PMNs to endothelium

LTC4, LTD4, LTE4 (identified as the SRS-A)
constrict extravascular smooth muscle
vasoconstrict
increase vascular permeability

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COMPLEMENT SYSTEM ACTIVATION

Innate system

- First line of defense
- Degranulation of mast cells/platelets
- Release of cytokines – initiation of inflammation
- Vascular changes
- Attraction of white blood cells

Factor	Source	Effect
Epidermal growth factor	Platelets, macrophages, endothelium	Chemotaxis, stimulates angiogenesis, wound contraction, fibroblast proliferation
Transforming growth factor-α	Macrophages, lymphocytes, keratinocytes	Stimulates keratinocyte and fibroblast proliferation, migration
Transforming growth factor-β	Platelets, lymphocytes, macrophages, endothelium, keratinocytes	Chemotactic, stimulates angiogenesis and fibroblast proliferation
Fibroblast growth factor	Macrophages, lymphocytes, endothelium, mast cells	Stimulates angiogenesis, keratinocyte and fibroblast proliferation, migration
Keratinocyte growth factor	Fibroblasts	Stimulates keratinocyte migration, differentiation, and proliferation
Tumor necrosis factor	Macrophages, lymphocytes, mast cells	Causes fever and corticotropin-releasing hormone secretion, suppresses appetite, stimulates acute-phase responses, stimulates macrophages, fibroblasts, angiogenesis
Interleukin-1	Macrophages, lymphocytes, mast cells	Causes fever, stimulates acute-phase response, stimulates neutrophils, macrophages, TNF, interferon release
Interleukin-2	Macrophages, lymphocytes, mast cells	Causes fever, stimulates macrophages, causes lymphocyte differentiation and proliferation
Interleukin-6	Macrophages, lymphocytes, mast cells	Causes fever, stimulates acute-phase response, up-regulation of Toll-like receptors
Interleukin-8	Macrophages, lymphocytes, mast cells	Chemotactic, stimulates neutrophils
Platelet-derived growth factor	Platelets, macrophages, endothelium	Chemotactic, stimulates fibroblasts, angiogenesis, wound contraction
Vascular endothelial growth factor	Keratinocytes, hypoxic cells	Stimulates angiogenesis
Endothelium-derived growth factor (nitric oxide)	Endothelium	Vasodilatation, antibacterial action, angiogenesis, apoptosis

Main cytokines involved in inflammation/wound healing

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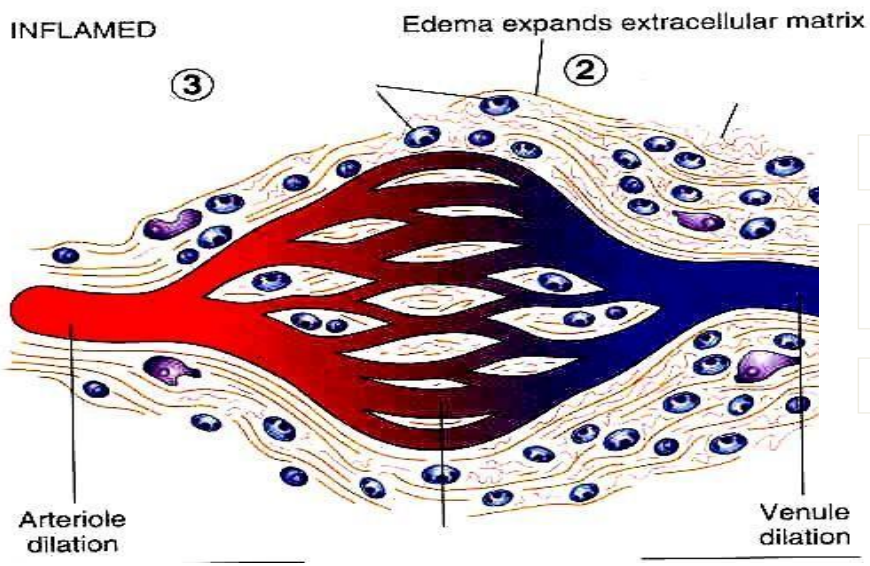
3 PROCESSES INVOLVED IN ACUTE INFLAMMATION



vascular dilatation	→	increased blood volume
structural changes in vessels allowing plasma to leak into tissues	→	increased volume of interstitial fluid
migration of leukocytes	→	increase in cellular components

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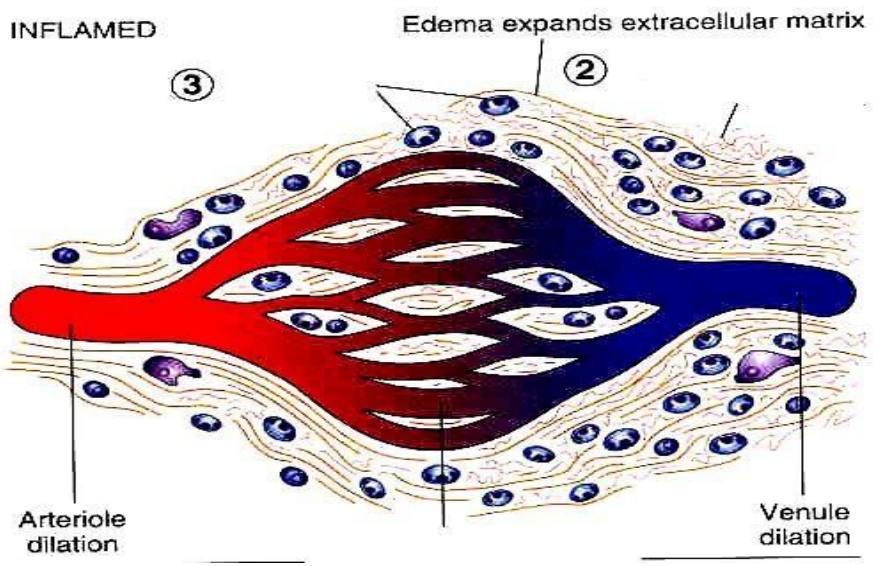
3 PROCESSES INVOLVED IN ACUTE INFLAMMATION



ALTERED BLOOD VESSEL CALIBER AND FLOW

VASODILATION → SLOWING OF FLOW → STASIS OF BLOOD CELLS →

LEUKOCYTE MARGINATION

INCREASED VASCULAR PERMEABILITY (LEAKAGE)

ESCAPE OF PROTEIN-RICH FLUID INTO INTERSTITIUM →

ALTERED OSMOTIC PRESSURE GRADIENT → FURTHER

OUTFLOW OF FLUID

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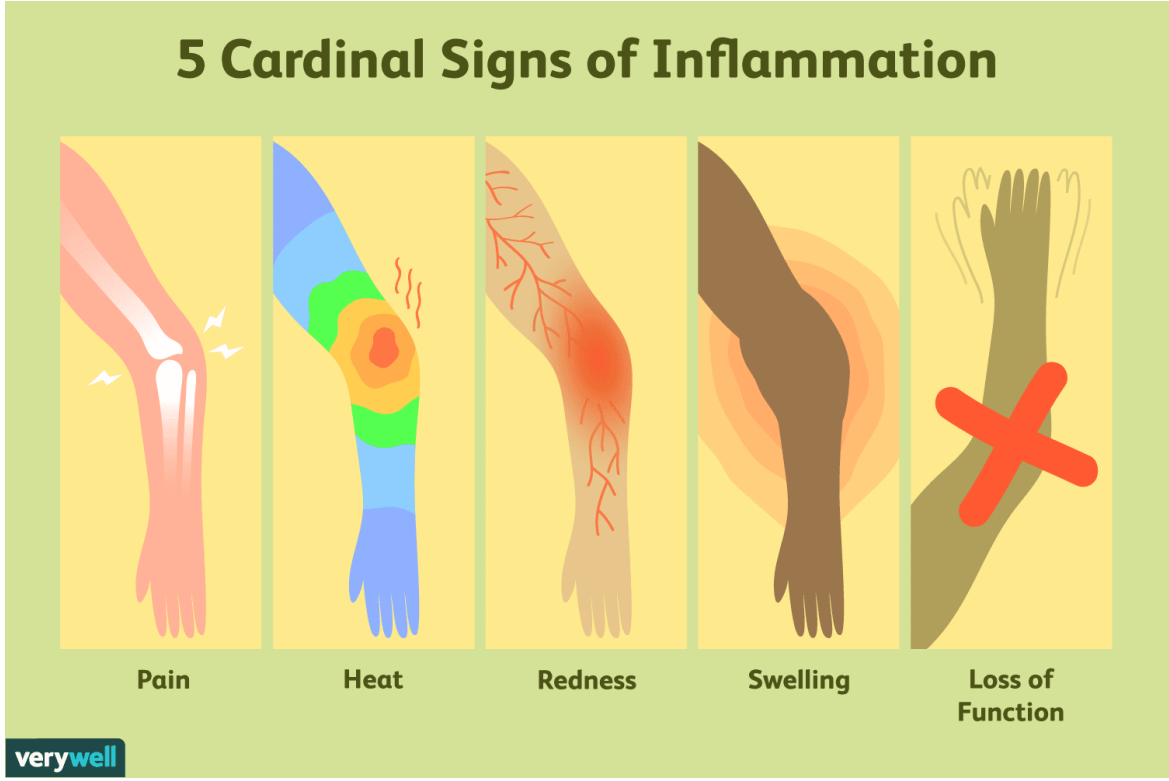
INFLAMMATION

The simplest definition of inflammation is best stated in Latin: calor, dolor, rubor, tumor

This assonant phrase refers to the heat (calor), pain (dolor), redness (rubor), and swelling (tumor) that characterize the clinical symptoms of inflammation as they were defined in the first century AD by the Roman scholar Celsus

Virchow added the fifth one loss of function (functio laesa)



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POSTOPERATIVE COURSE AFTER ORAL SURGERY:

Inflammation

- Pain
- Swelling
- Redness

Non-functioning muscles

Trismus

Illness and pyrexia

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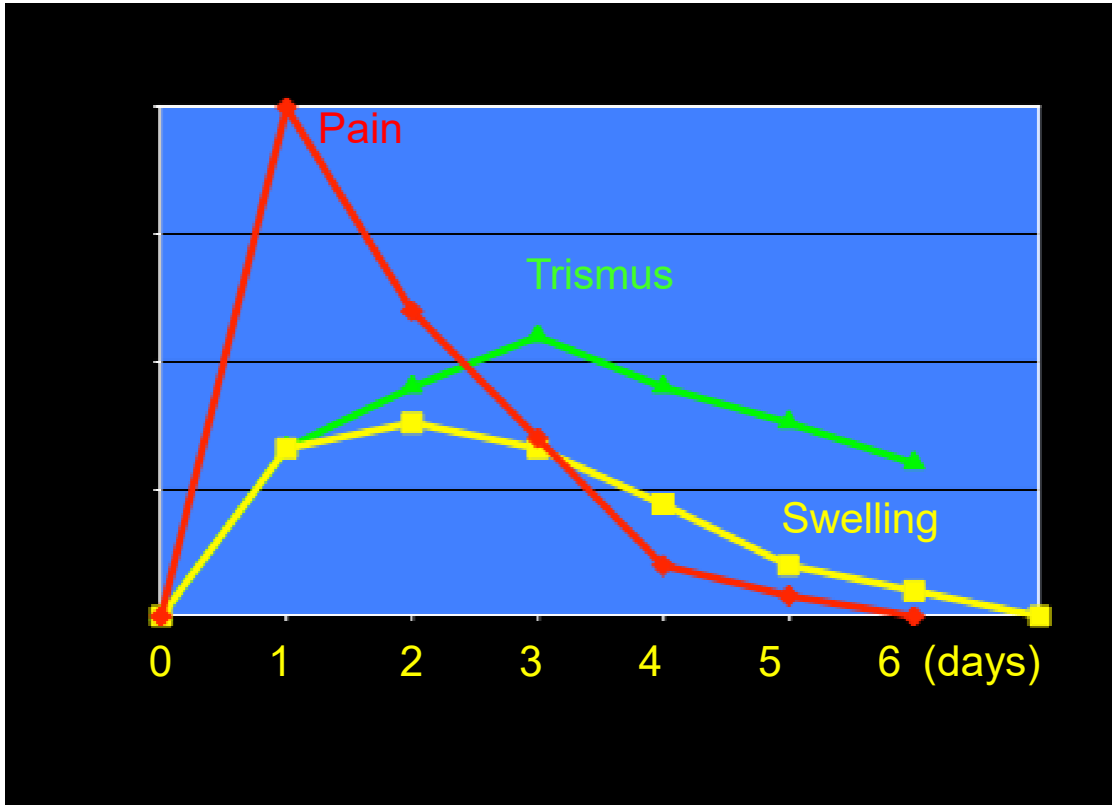
POSTOPERATIVE COURSE EVOLUTION AFTER ORAL SURGERY



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POSTOPERATIVE PAIN

Difficulty

- Extent of surgery
- Flap
- Bone removal
- Maxilla vs. mandible
- Anterior vs. posterior

Individual sensitivity

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POSTOPERATIVE PAIN MANAGEMENT

NSAIDs

Tissue Damage

Exposure of cell membrane phospholipids

phospholipase

Arachidonic Acid Formation

Cox inhibitors (NSAIDs)

Selective 2 inhibitors
Selecoxib
Rofecoxib

1,2 inhibitors
Aspirin
Ibuprofen

cyclooxygenase (COX) 1,2

Lipooxygenase

Prostaglandins
(PGD_2, PGE_2, PGF_2) vasodilation
Prostacyclin (PGI_2) vasodilation, antiplatelet aggregation
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Leukotrienes → Vasoconstriction
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Incr. permeability

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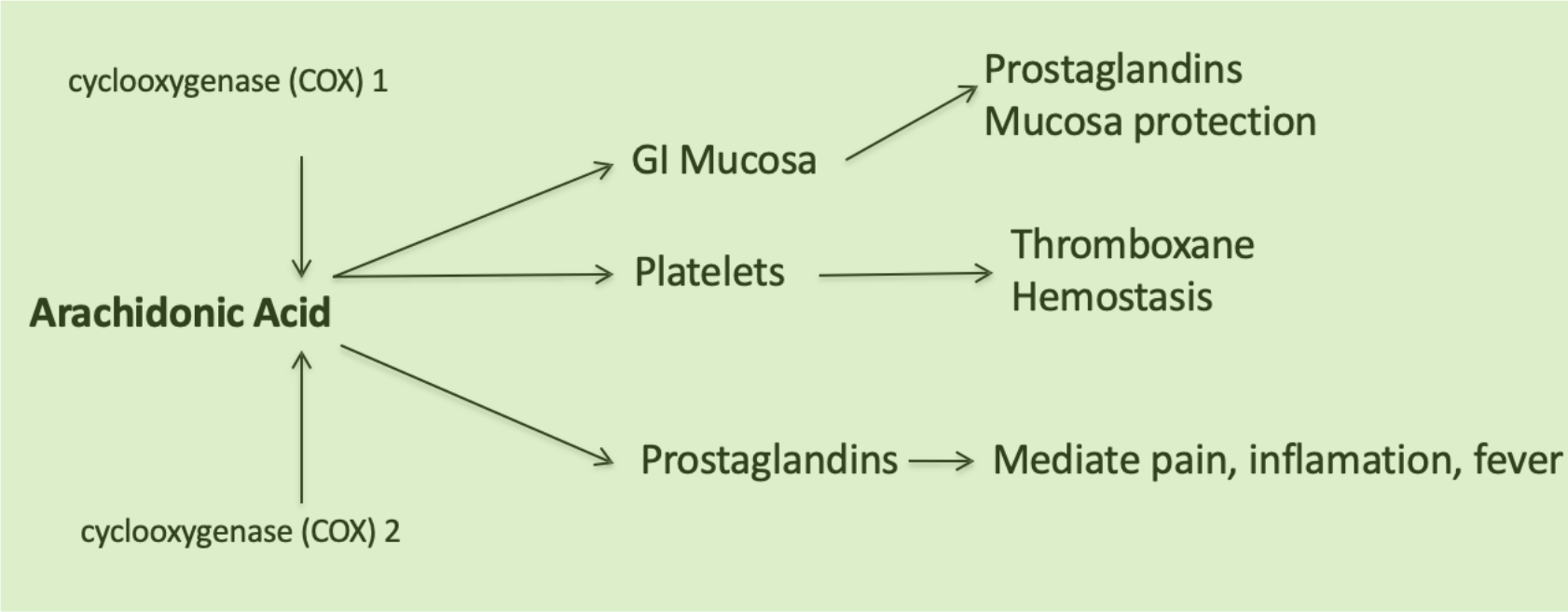
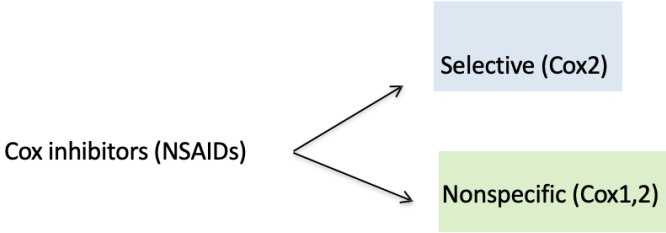
POSTOPERATIVE PAIN MANAGEMENT





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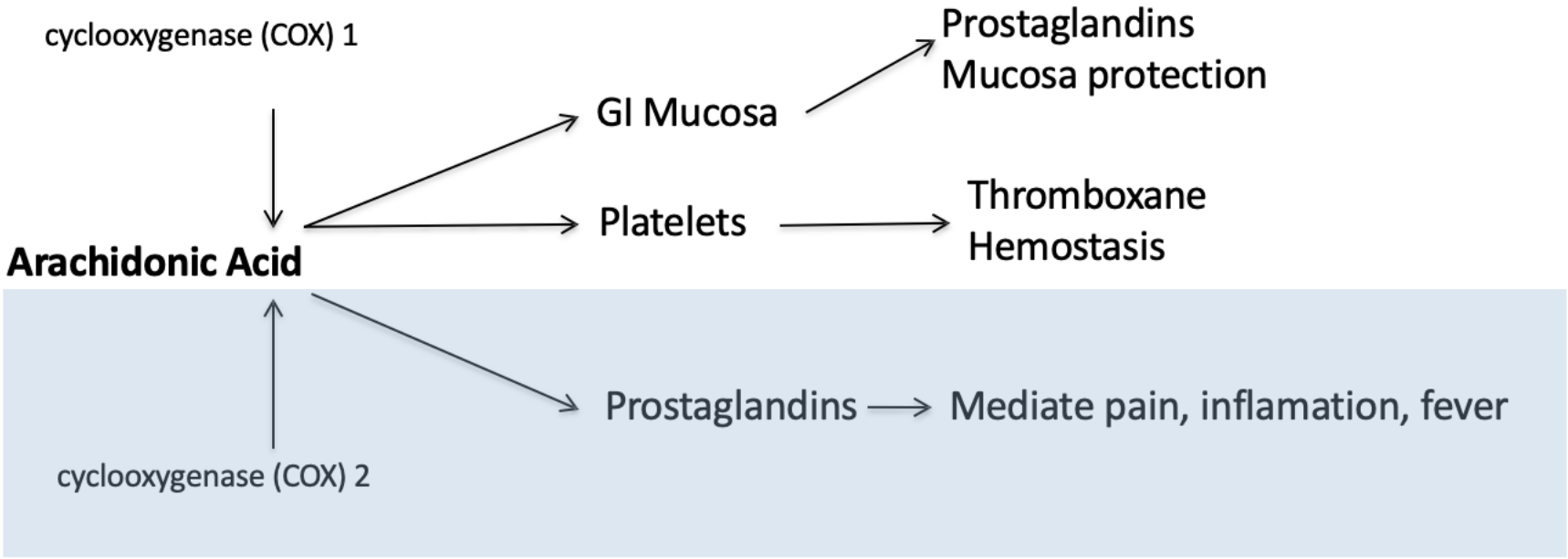
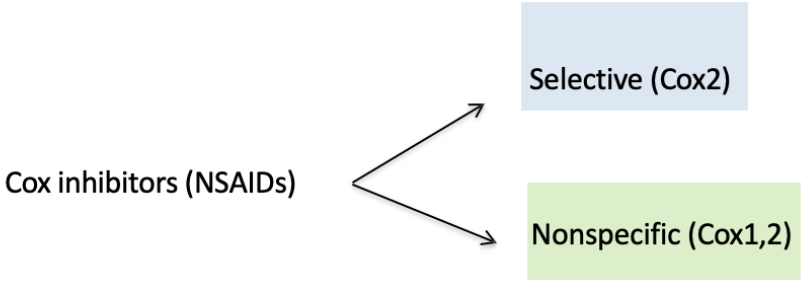
POSTOPERATIVE PAIN MANAGEMENT





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POSTOPERATIVE PAIN MANAGEMENT

Opioids

Mechanism of action: Bind to specific receptors either in CNS, the spinal cord or other organs

The analgesic effect is caused by the actual blocking of neurotransmission at the presynaptic level in efferent pain pathways

The above is augmented by the euphoric effect they induce by acting on the dopamine system of the brain modulating the individual's reaction/feeling of pain

Delta (δ)	Brain Peripheral neurons	Analgesia Antidepressant effects Dependence
Kappa (κ)	Brain Spinal cord Peripheral sensory neurons	Analgesia Sedation Miosis Dysphoria
Mu (μ)	Brain Spinal cord Peripheral neurons Intestinal tract	Resp. depression Analgesia Euphoria Reduced GI mobility Dependence
Nociceptin	Brain Spinal cord	Anxiety Depression

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POSTOPERATIVE PAIN MANAGEMENT

NSAIDs

- Analgesia
- Anti-inflammatory
- Antipyretic
- GI upset-vomiting
- Increased bleeding
- Renal damage

Opioids

- Analgesia
- Respiratory depression
- Antitussive
- Mood alteration
- Nausea
- Dependence
- Constipation

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POSTOPERATIVE PAIN MANAGEMENT

NSAIDs

- Allergy
- Gastric ulcers
- Renal disease
- Coagulation impairment
- Severe asthma

Opioids

- Severe COPD
- Dependency
- Severe inflammatory bowel disease
- Concurrent use of CNS depressants

  - Alcohol

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POSTOPERATIVE SWELLING

Individual reaction

Degree of trauma

- Flap raised
- Time to complete surgery
- Presence of infection previously

Suturing technique “tightness” of closure

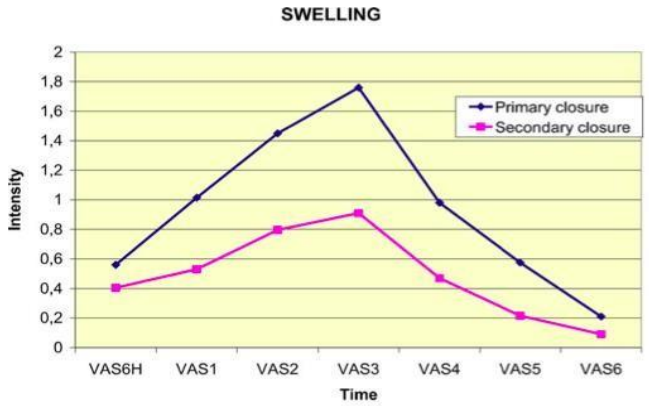
Steroid use

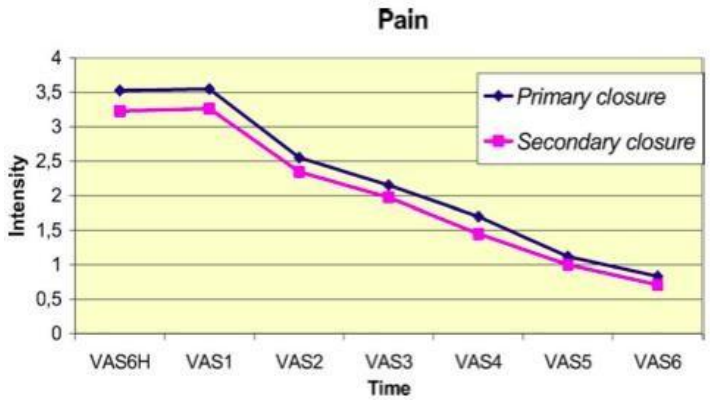
International Journal of Oral and Maxillofacial Surgery
Volume 34, Issue 1, January 2005, Pages 52-57

Clinical Paper
Oral Surgery

Primary and secondary closure of the surgical wound after removal of impacted mandibular third molars: a comparative study

D. Pasqualini [1], N. Cocero [1], A. Castella [2], L. Mela [3], P. Bracco [1]





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POSTOPERATIVE SWELLING

- Individual reaction
- Degree of trauma
  - Flap raised
  - Time to complete surgery
  - Presence of infection previously
- Suturing technique “tightness” of closure
- Steroid use

Experience in third molar surgery: an update

W. Jerjes, T. Upile, F. Nhembe, D. Gudka, P. Shah, S. Abbas, E. McCarthy, S. Patel, J. Mahil & C. Hopper

British Dental Journal **209**, E1 (2010) | Cite this article

5928 Accesses | **33** Citations | **261** Altmetric | Metrics

Method In this study, 3,236 patients underwent surgical removal of impacted third molars. All patients included in the study were reviewed and the various postoperative complications were recorded and statistically compared to the surgeon's grade. Patients' demographics and pre-operative radiographic findings were also noted.

Results The surgical procedures were performed by seven specialists and 12 residents. In the group of patients treated by the residents, the incidence of postoperative complications was found to be significant with regards to trismus, infection, alveolar osteitis and paraesthesia of the lingual and inferior alveolar nerves. In the group of patients treated by specialists, the incidence of postoperative bleeding was found to be statistically significant.

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POSTOPERATIVE SWELLING

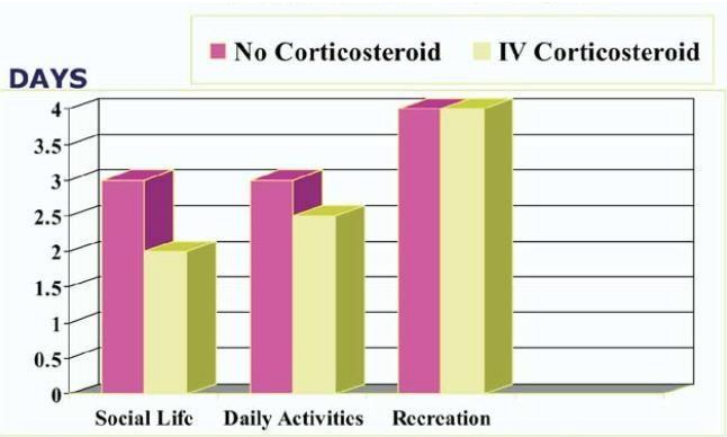
Steroid Use

Tiwana et al. IV Corticosteroids With Third Molar Surgery.
J Oral Maxillofac Surg. 2005 Jan;63(1):55-62

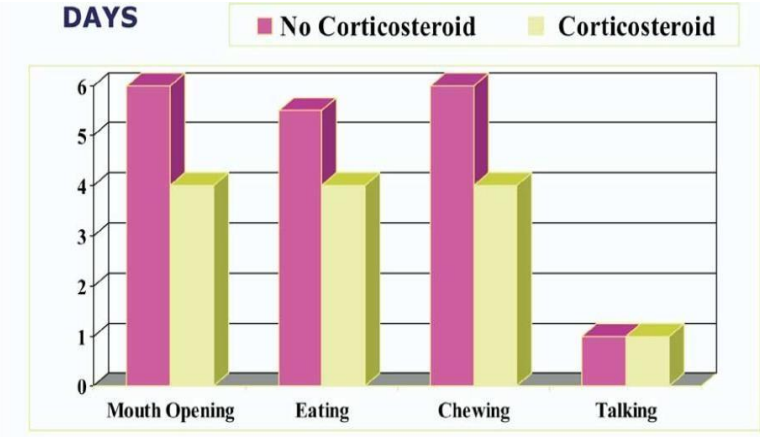
J Dent Anesth Pain Med. 2020 Oct; 20(5): 281–291.
Published online 2020 Oct 30. doi: 10.17245/jdapm.2020.20.5.281
PMCID: PMC7644360
PMID: 33195806

Do corticosteroids reduce postoperative pain following third molar intervention?

Chollathit Sugragan, Nattapong Sirintawat, Sirichai Kiattavornchareon, Lee Kian Khoo, Kumar KC, and Natthamet Wongsirichat



Differences not statistically significant P>0.05



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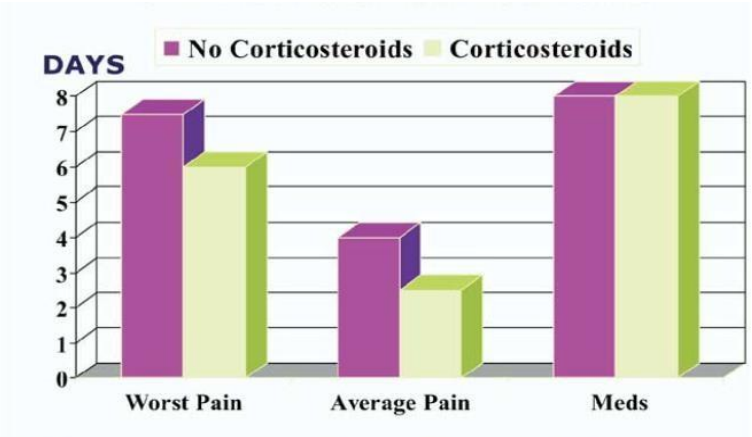

Differences not statistically significant P>0.05

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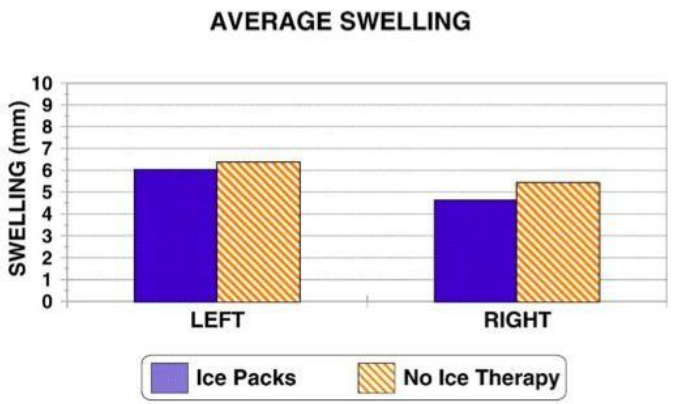
POSTOPERATIVE SWELLING

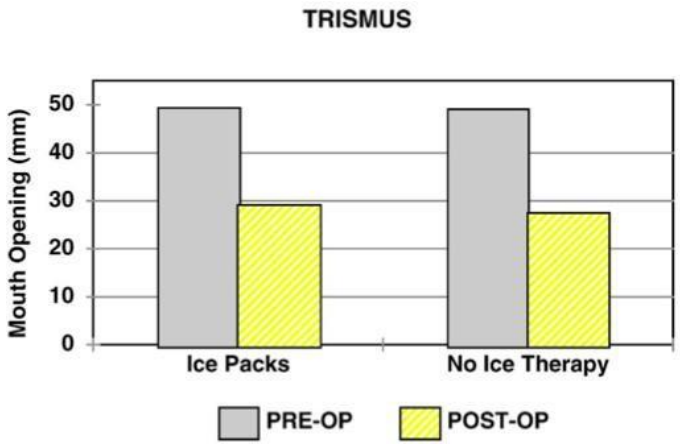
Use of ice packs

Van der Westhuijzen AJ, Becker PJ, Morkel J, Roelse JA.
A Randomized Observer Blind Comparison of Bilateral Facial Ice Pack Therapy With No Ice Therapy Following Third Molar Surgery
Int. J. Oral Maxillofac. Surg. 2005; 34: 281–286.

Table 2. Patient's perception of postoperative control of symptoms

	Frequency (*N*)		
	Ice therapy	No ice therapy	Total
Excellent	18	8	26
Good	10	19	29
Moderate	1	3	4
Total	29	30	59





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POSTOPERATIVE SWELLING

Swelling around the mouth, eyes, and cheeks is a normal reaction to the surgery and usually takes 2-3 days to fully develop

To help prevent swelling:

•Apply ice packs to the cheeks for 20 minutes on and 20 minutes off. Keep your head elevated for first 48 hours after surgery. Use crushed ice in a ziplock bag or a bag of frozen peas bag wrapped in a moist cloth to prevent frostbite

•After the first 3 days, you can begin using a heating pad or moist heat for relief of swelling, bruising, and stiffness of the jaw. Apply heat 4 times/day for 30 minutes time intervals

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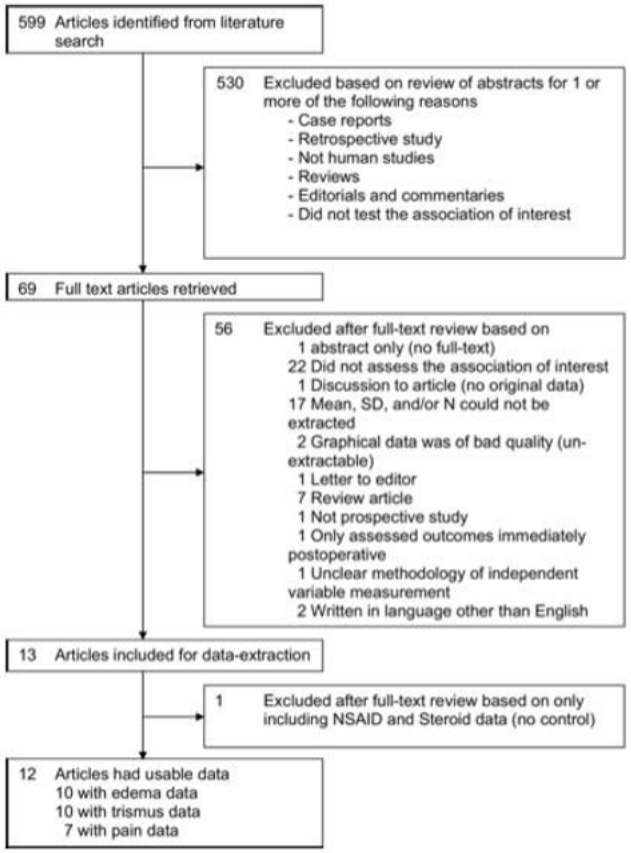
POSTOPERATIVE USE OF CORTICOSTEROIDS

Corticosteroids Reduce Postoperative Morbidity After Third Molar Surgery:
A Systematic Review and Meta-Analysis

Michael R. Markiewicz, DDS, MPH,* Mark F. Brady, BS,† Eric L. Ding, ScD,‡ and Thomas B. Dodson, DMD, MPH§

J Oral Maxillofac Surg 66:1881-1894, 2008

The findings of this study suggest that perioperative administration of corticosteroids produces a mild to moderate reduction in edema and improvement in range of motion after M3 removal.



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POSTOPERATIVE USE OF CORTICOSTEROIDS

ABSOLUTE

- TB: active or healed
- Ocular herpes simplex
- Primary glaucoma
- Acute psychosis
- Allergy

RELATIVE

- Diverticulitis
- Recent intestinal anastomosis
- Peptic ulcer disease
- Cushing's syndrome
- Renal insufficiency
- Hypertension
- Thromboembolic tendencies or thrombophlebitis
- Osteoporosis
- Diabetes
- Psychotic tendencies Acute or chronic infections
  First trimester pregnancy

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POSTOPERATIVE TRISMUS

Trismus - limits mouth opening in an attempt to prevent additional trauma or pain after third molar surgery

Muscles of mastication (closers) unable to function due to inflammation

- degree of trismus correlates with degree of trauma

Differential Diagnosis:

Hematoma

TMJ disk displacement

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POSTOPERATIVE COURSE - OTHER

Pyrexia

Dysphagia

Inability to work (2-5 days)

Severe inflammatory reaction

- Lack of food intake
- Lack of fluid intake
- Drug side effects

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THANK YOU

Maria Franco Echevarria, MD, DMD, MSc, FRCDC

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